Management and Entrepreneurship

IOT-Based AQI Meter System for Real-Time Data Analysis and Decision Making

Project Proposal

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ABSTRACT:

The project is to develop a system for measuring real-time indoor and outdoor air quality using the IoT-based Air Quality Index Monitoring System. On collected data analysis, users will be able to make better decisions. This system also encompasses the management and entrepreneurship aspect in making the system market-ready, scalable, and sustainable. The system includes IoT-enabled sensors (PM2.5 , PM10, MQ-135 and MQ-5) integrated with ESP32, ESP8266 and Arduino to gather, transmit, and store air-quality data on cloud platforms for further analyses.

PROBLEM STATEMENT:

It is going to be an IoT-based project to develop the AQI real-time monitoring system which will monitor air pollutants related to health issues and environment degradation hampered by urbanization and industrialization. The project will identify important pollutants, study their effects on health and the environment, and enable a system to monitor the pollutants continuously. The concern for this solution into the commercial realm includes urban planning, public health management, and industrial compliance; and the primary beneficiary public are environmental agencies, local governments, industries, and also the general public.

OBJECTIVES:

The objective of the research is to create a real-time IoT-based air quality monitoring solution for indoor and outdoor purposes, implement cloud-based analytics into it, and assess for performance, accuracy, and user-friendliness, identifying areas for entrepreneurial opportunities to commercialize the product for homes, industries, and urban areas.

PROPOSED SOLUTIONS:

The proposed solutions consist of hardware components, software components , real-time monitoring, alerts and notifications.

Hardware Components:

The package consists of an Arduino,ESP32,ESP8266, MQ series gas sensor(MQ-135,MQ-5), PM2.5 and PM10 sensors, battery pack, power adapter and charger setup, weatherproof housing, WiFi/Bluetooth modules, board, and connecting wires.

Software Components:

It utilizes numerous tools and languages including Blynk App, Firebase, Android Studio, MATLAB, Python, Arduino framework, ECLIPSE IDE, code for firewall coding, things speak, MQTT protocols, AWS IoT, Adafruit I/O, GitHub, Auth Mechanisms, Grafana, Excel, HTTP/HTTPS, sensors, and WiFi Libraries.

Real-time monitoring:

This available data will be user friendly processed and shown instantaneously through the QM system. The QM uses standardized environmental parameters for AQI calculation.

Alerts and Notifications:

Threshold alerts will inform users hazardous levels of air quality.

MANAGEMENT AND ENTERPRENUERSHIP SCOPE:

It is anticipated that the market for global air quality monitoring will enjoy growth due to increasing awareness about pollution. House-qualified, companies, government institutions, and health care institutes will form target customers. The product will follow a hybrid model by selling IoT-based AQI devices directly, providing subscription services, and including industries or municipalities in collaborative programs. Hardware procurement, software development, and cloud-architecture in the initial cost will incur every scalability issue but lower production costs. The revenue stream is to be through sales of devices, subscription fees, and custom solutions to businesses.

BENEFITS AND IMPACT:

Real-time air quality data helps improve the quality of the air; reduce exposure to polluted air; promote a healthier lifestyle; and create business opportunities. It can also help save on healthcare costs related to diseases caused by pollution and boost overall economic growth.

CONCLUSION:

In conclusion, the IOT-based AQI Monitoring System Technology Management and entrepreneurship opened up the possibility for a low-cost, scalable, and impactful solution in air quality monitoring. Improvements in real-time data use and analysis contribute to a cleaner and healthier environment while creating business opportunities and sustainability.